UNIVERSAL CLOSEDNESS AND PROPERNESS ARE PRESERVED WHEN THE TARGET IS SEPARATED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-algebraic-stacks-morphisms/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

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Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The statement that the graph of a morphism into a separated stack is proper. Invoked inside the counted proof (source lines 8677-8678). Makes the first factor of the factorization universally closed.
- The base change permanence of universal closedness. Invoked inside the counted proof (source lines 8679-8682). Makes the projection universally closed.
- The composition permanence of universal closedness. Invoked inside the counted proof (source lines 8683-8685). Combines the two factors.
- The separatedness and finite type permanence statements. Invoked at the end of the counted proof (source lines 8686-8690). Upgrades the conclusion from universally closed to proper.

Source reference. Original lemma lemma-base-change-universally-closed: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 2726–2726. Navigation pointer only: the counted text refers to the Stacks item lemma-base-change-universally-closed, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-compose-after-separated: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 929–929. Navigation pointer only: the counted text refers to the Stacks item lemma-compose-after-separated, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

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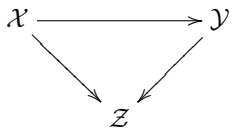
Source reference. Original lemma lemma-composition-universally-closed: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 2736–2736. Navigation pointer only: the counted text refers to the Stacks item lemma-composition-universally-closed, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-finite-type-permanence: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 3608–3608. Navigation pointer only: the counted text refers to the Stacks item lemma-finite-type-permanence, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-quasi-compact-permanence: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 2044–2044. Navigation pointer only: the counted text refers to the Stacks item lemma-quasi-compact-permanence, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-semi-diagonal: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 808–808. Navigation pointer only: the counted text refers to the Stacks item lemma-semi-diagonal, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Consider a commutative diagram*



of algebraic stacks.

(1) *If $\mathcal{X} \to \mathcal{Z}$ is universally closed and $\mathcal{Y} \to \mathcal{Z}$ is separated, then the morphism $\mathcal{X} \to \mathcal{Y}$ is universally closed. In particular, the image of $|\mathcal{X}|$ in $|\mathcal{Y}|$ is closed.*
(2) *If $\mathcal{X} \to \mathcal{Z}$ is proper and $\mathcal{Y} \to \mathcal{Z}$ is separated, then the morphism $\mathcal{X} \to \mathcal{Y}$ is proper.*

Proof. Assume $\mathcal{X} \to \mathcal{Z}$ is universally closed and $\mathcal{Y} \to \mathcal{Z}$ is separated. We factor the morphism as $\mathcal{X} \to \mathcal{X} \times_{\mathcal{Z}} \mathcal{Y} \to \mathcal{Y}$. The first morphism is proper (Lemma lemma-semi-diagonal) hence universally closed. The projection $\mathcal{X} \times_{\mathcal{Z}} \mathcal{Y} \to \mathcal{Y}$ is the base change of a universally closed morphism and hence universally closed, see Lemma lemma-base-change-universally-closed. Thus $\mathcal{X} \to \mathcal{Y}$ is universally closed as the composition of universally closed morphisms, see Lemma lemma-composition-universally-closed. This proves (1). To deduce (2) combine (1) with Lemmas lemma-compose-after-separated, lemma-quasi-compact-permanence, and lemma-finite-type-permanence. □